

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

PRACTICAL 7(Task)

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Q.1

Area (sq.ft.) Price (Lakhs)

600 32

800 40

1000 49

1200 58

1400 68

1600 79

Find the regression equation.

Predict the price of a 1500 sq.ft. house.

Predict the price of a 2500 sq.ft. house.

Code:-

```
import pandas as pd
from sklearn.linear_model import LinearRegression
# Load dataset
df = pd.read_csv("land.csv")
# Display dataset
print(df)
# Independent Variable
X = df[["Area"]]
# Dependent Variable
y = df["Price"]
# Train model
model = LinearRegression()
model.fit(X, y)
# Regression Equation
slope = model.coef_[0]
intercept = model.intercept_
print("\nRegression Equation:")
print(f"Price = {slope:.4f} * Area + {intercept:.4f}")
# Predictions
price_1500 = model.predict([[1500]])
price_2500 = model.predict([[2500]])
print("\nPredicted Price for 1500 sq.ft.:", round(price_1500[0], 2), "Lakhs")
```

```
print("Predicted Price for 2500 sq.ft.:", round(price_2500[0], 2), "Lakhs")
```

Output:-

```
Area  Price
0    600    32
1    800    40
2   1000    49
3   1200    58
4   1400    68
5   1600    79
```

Regression Equation:

Price = 0.0469 * Area + 2.7905

Predicted Price for 1500 sq.ft.: 73.08 Lakhs

Predicted Price for 2500 sq.ft.: 119.93 Lakhs

Q.2 A company has collected the following employee data:

- Employee ID
- Employee Age
- Years of Experience
- Salary

You need to develop a Linear Regression model to predict an employee's salary.

Tasks

1. Identify the independent variable (X) and the dependent variable (Y).
2. Explain why you selected these variables.
3. Create a Pandas DataFrame using the given data.
4. Split the dataset into training and testing sets.
5. Train a Linear Regression model.
6. Predict the salary for an employee having 7 years of experience.
7. Display:
 - o Slope (Coefficient)
 - o Intercept
 - o R2 Score
8. Plot the regression line.

Code:-

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
# Create DataFrame
data = {
    "Employee_ID": [101,102,103,104,105,106,107,108],
    "Age": [22,25,28,30,35,40,45,50],
    "Experience": [1,2,3,5,6,8,10,12],
    "Salary": [25000,30000,35000,45000,50000,65000,80000,95000]
}
df = pd.DataFrame(data)
```

```

print(df)
# Independent and Dependent Variables
X = df[['Experience']]
y = df['Salary']
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Train Model
model = LinearRegression()
model.fit(X_train, y_train)
# Predict Salary for 7 years of experience
salary = model.predict([[7]])
print("Predicted Salary for 7 Years Experience:", salary[0])
# Display Coefficient, Intercept and R2 Score
print("Slope (Coefficient):", model.coef_[0])
print("Intercept:", model.intercept_)
y_pred = model.predict(X_test)
print("R2 Score:", r2_score(y_test, y_pred))
# Plot Regression Line
plt.scatter(X, y, color='blue')
plt.plot(X, model.predict(X), color='red')
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Employee Salary Prediction")
plt.show()
Output:-

```

	Employee_ID	Age	Experience	Salary
0	101	22	1	25000
1	102	25	2	30000
2	103	28	3	35000
3	104	30	5	45000
4	105	35	6	50000
5	106	40	8	65000
6	107	45	10	80000
7	108	50	12	95000

Predicted Salary for 7 Years Experience: 60374.280230326294
 Slope (Coefficient): 6449.136276391556
 Intercept: 15230.3262955854
 R2 Score: 0.9888538938329816

